

Python_Grundlagen_Test

December 4, 2020

1 Practicing Python Basics (iPython Notebook)

1.0.1 Solve the following five exercises - each proper solution is worth 1 point.

Exercise 1: Take the following list and write a program that prints out all the elements of the list that are less than 5.

```
[1]: a = [1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89]
      print([i for i in a if i < 5],
            "<- 1-line solution to print elements that are less than 5")
```

```
[1, 1, 2, 3] <- 1-line solution to print elements that are less than 5
```

Exercise 2: Take the following two lists and write a program that returns a list that contains all elements of the lists (without duplicates). Make sure your program works on two lists of different sizes. Moreover, try to find a 1-line-solution (using sets).

```
[2]: import random
      a = random.sample(range(1, 100), 20)
      b = random.sample(range(1, 100), 25)
      print(list(set(a + b)),
            "<- 1-line solution to make a list of two lists without repetition")
```

```
[3, 4, 13, 14, 15, 17, 19, 24, 27, 29, 30, 37, 40, 41, 42, 45, 47, 51, 53, 54,
57, 60, 68, 70, 72, 75, 76, 77, 79, 80, 81, 83, 85, 87, 88, 89, 94, 96, 97, 98,
99] <- 1-line solution to make a list of two lists without repetition
```

Exercise 3: Implement a function that takes as input a list of numbers, and returns the largest one. Do this without using the Python max() function!

```
[3]: def largest_number(my_list): # wanted Function
      my_list.sort()
      return my_list[-1]

      print(a, "<- Original list")
      print((largest_number(a)), "<- Largest number in the original list")
```

```
[81, 77, 4, 13, 68, 80, 42, 85, 97, 47, 51, 72, 37, 60, 88, 45, 24, 98, 27, 57]
<- Original list
98 <- Largest number in the original list
```

Exercise 4: Write a Python program to concatenate following dictionaries to create a new one.

```
[4]: dic1={1:10, 2:20}
     dic2={3:30, 4:40}
     dic3={5:50, 6:60}
     dic4 = {}
     for d in (dic1, dic2, dic3): dic4.update(d)
     print(dic4, "<- Concatenated new dictionary")
```

```
{1: 10, 2: 20, 3: 30, 4: 40, 5: 50, 6: 60} <- Concatenated new dictionary
```

Exercise 5: Generate a random number between 1 and 15 (including 1 and 15). Ask the user to guess the number, then tell him/her whether he/she guessed too low, too high, or exactly right. Remark: Import and use the random library.

```
[5]: # generate a random number
     import random
     number = random.randint(1,15)
     user_input = int(input("If you guess the random number between 1 and 15"
                           "you are the winner!\nPlease enter a number: "))
     if number == user_input:
         print("We have a winner, you have guessed the correct number!!!")
     elif number < user_input:
         print("Unfortunately you guessed to high,"
               f" the winning number was {number} but you chose {user_input} :(")
     else:
         print("Unfortunately you guessed to low,"
               f" the winning number was {number} but you chose {user_input} :(")
```

```
If you guess the random number between 1 and 15you are the winner!
```

```
Please enter a number: 5
```

```
Unfortunately you guessed to low, the winning number was 11 but you chose 5 :(
```